Adaptive and User-Centered Filter Coffee Machine Design

Presented by Bahar Kabayuka

Key Design Milestones (Timeline / Process Visualization)

1 – Problem Identification

Most home drip coffee machines provide a standard brewing experience and do not adapt well to different user preferences. Many machines are designed mainly for one type of container and do not easily support different serving sizes such as a cup, mug, or carafe. This creates a gap between user expectations and product functionality.

2 – User Research

After asking several users about their experiences with coffee machines, a number of needs were identified:

- different coffee strength levels (strong, medium, soft)
- different serving sizes (cup, mug, carafe)
- the ability to save a favorite taste setting
- easier cleaning and maintenance
- a removable water tank
- the ability to see the water level from the outside

Many users mentioned that being able to completely remove the water tank would make filling and cleaning easier.

3 – Concept Development

Based on these insights, the concept focused on creating a personalized coffee experience. The design includes several features:

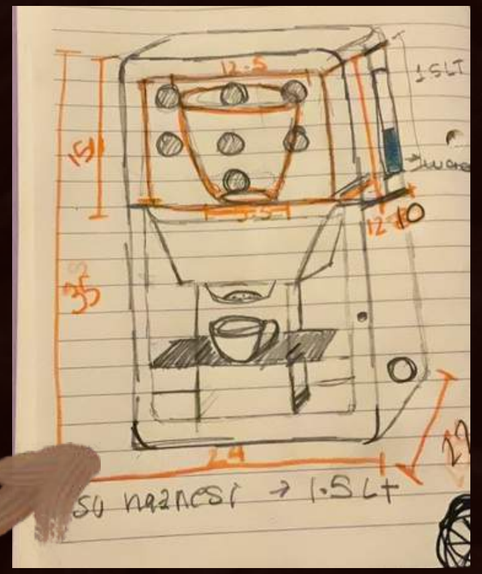
- three brewing modes (Cup, Mug, Carafe)
- an adjustable platform for different container heights
- a favorite taste memory option
- a visible water level indicator
- a fully removable water tank

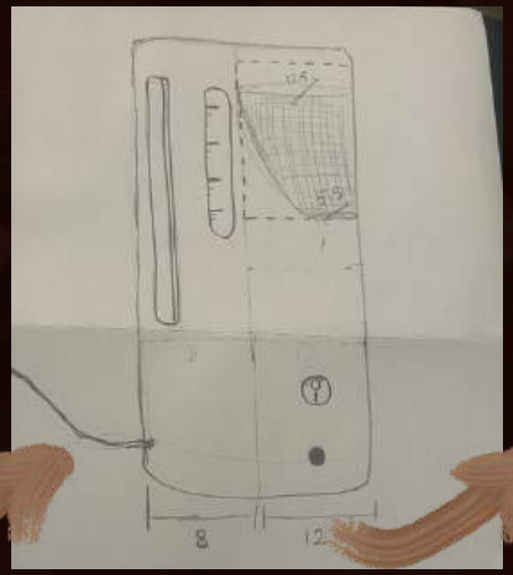
To make the product more interactive and enjoyable, a chalkboard surface was added so users can write notes or messages.

my sketches

Prompt
Refine the coffee machine design by adding a circular illuminated button interface on the upper front panel. The buttons should represent coffee options and strength levels. Maintain the black matte surface with soft golden lighting details for a premium product design.

Prompt
Modernize it a bit more; the previous one was too classic.





I created a drawing showing the interior design along with its measurements, and then asked the chat application to visualize it.

Prompt
It's too flat, add some rounded edges, add gold details, and make it a chalkboard on the side. Add buttons to raise and lower the elevator system.

Process

Chalkboard Note Area
The machine includes a small chalkboard surface where users can write notes or draw.
This feature adds a fun and playful interaction to the product.

Self-contained Filter Basket
The machine includes a built-in filter basket, so users do not need a paper filter.
After brewing, the basket slides out easily for cleaning and removing coffee grounds.

Aroma Selection
Users can choose between soft, medium, or strong coffee taste. The machine adjusts brewing conditions accordingly

Touchscreen Control Pael
A simple touchscree interface allows users to select aroma level and brewing settings easily.

Pull-out Water Tank
The side sliding water tank makes filling and cleaning easier.

Water Level Indicator
The illuminated side indicator shows the water level and machine activity.

Easy Cleaning Structure
Removable parts allow quick and simple cleaning.

Adjustable Elevator System
The machine includes an adjustable elevator platform, allowing users to place either a glass carafe or a single cup depending on their preference.

Sliding Silicone Brewing Base
The brewing area has a silicone base that slides out, providing stability and preventing cups or carafes from slipping during brewing.

Final Process

1. KEY WORDS

- Adjustable elevator brewing system
- Built-in filter basket
- Touchscreen control panel
- Aroma selection
- Pull-out water tank
- Even water distribution (shower head)
- Splash reduction
- Sliding silicone brewing base
- Glass carafe compatibility
- Chalkboard interaction surface

2. Selected Effective Questions

- Q1
- How can the coffee machine adapt to both a glass carafe and a single cup?
- Q2
- How can the brewing system reduce coffee splashing during pouring?
- Q3
- How can the user customize coffee taste and aroma easily?
- Q4
- How can the design make cleaning and maintenance easier?
- Q5
- How can the product create a more playful and interactive user experience?

AI Role Labels

Explainer
AI provided explanations about the coffee brewing process, including water temperature, aroma formation, and user experience considerations.

Reframer
AI suggested thinking about the product not only as a coffee machine but also as a system that can adapt to different user preferences.

Expander
AI proposed additional ideas such as lighting elements, improving the brewing system, and enhancing user interaction.

Critical
AI raised questions about functionality, especially regarding splash reduction, usability, and the brewing mechanism.

AI Response Summary
During the design process, AI mainly supported the research and information-gathering stages. It provided useful information about coffee brewing, aroma development, and user experience. However, AI sometimes struggled to fully understand the visual instructions I provided and could not always evaluate the design from a highly technical perspective. Therefore, AI was mainly used as a tool for information support and idea exploration.

My Reflections
Working with AI helped me explore different ideas and understand some aspects of coffee brewing and user interaction. It was especially useful during the research phase. However, when it came to visual and technical design decisions, most of the decisions were made by me. AI acted more as a supportive tool rather than the main design driver.

Design Decisions
Based on research and AI-supported discussions, I made the following design decisions:

- Designing a brewing system that reduces coffee splashing
- Using lighting elements to provide visual feedback to the user
- Creating a modern and minimal interface
- Allowing the machine to work with both a single cup and a glass carafe through an adjustable system
- Designing pull-out components such as the water tank and filter area for easier use
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Accepted, Rejected, and Transformed Ideas

Accepted

- Adjustable elevator brewing system
- Built-in filter basket
- Silicone anti-slip brewing base
- Lighting elements
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Rejected

- Fixed brewing height
- Complex control systems that reduce usability
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Transformed
The design evolved from a simple drip coffee machine into a more flexible and user-oriented brewing system.